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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 18

B.Tech. (ECE/Electronics Engg) (2012 to 2017)
(Sem.–4)

ELECTROMAGNETICS AND ANTENNAS

Subject Code : BTEC-403
M.Code : 57595

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

Answer briefly :

1) Calculate the radiation resistance of $\lambda/10$ wire dipole in free space?

2) Define Antenna beam width.

3) What are the disadvantages of ground wave propagation?

4) What is retarded potential?

5) State field equivalence principle.

6) Define Surface Impedance.

7) Differentiate between TE, TM and TEM mode.

8) What is Distortion Less line?

9) Define Polarization.

10) What are smith charts? Briefly explain.

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SECTION-B

11) What is aperture antenna? Explain its different types.

12) Explain the working of parabolic reflector antenna.

13) Explain Duct Propagation in detail.

14) A rectangular waveguide with dimensions 4cm × 2cm operates at 10 GHz. Find f_c and λ_c of TE_{10} mode.

15) Discuss the wave propagation characteristics in good dielectric.

SECTION-C

16) Explain the concept of radiation in single wire, two wire and dipole antenna.

17) For an array of n isotropic point sources radiating in end fire array, derive and obtain the maxima and minima direction for major and minor lobes. Sketch the pattern.

18) Give the transmission line analogy for waveguides. Also draw the equivalent transmission line circuit representation for TE and TM waves.

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